

HEUKSANDO, Korea, Republic of

WMO: 471690

Lat: 34.6872N

Lon: 125.4511E

Elev: 78

StdP: 100.39

Time Zone: 9.00 (E09)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.2	-8.3	1.9	0.4	-7.2	2.1	0.9	15.4	0.9	14.6	0.9	10.0	340	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.3	29.5	26.6	28.3	25.9	27.1	25.1	27.0	29.0	26.2	27.8	25.5	26.8	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.2	28.2	25.7	21.2	27.4	25.0	20.3	26.6	86.0	29.1	82.3	27.9	79.0	26.9	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	12.8	11.4	DB	-3.8	30.9	1.9	1.3	-5.2	31.8	-6.3	32.6	-7.4	33.4	-8.8	34.3	(4)
			WB	-5.0	27.8	1.6	1.5	-6.1	28.8	-7.1	29.7	-8.0	30.5	-9.2	31.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	3.4	3.8	6.6	10.8	15.5	19.5	23.1	25.0	21.6	17.1	11.5	5.9
	DBStd	7.79	2.74	2.86	2.64	2.44	2.20	1.83	2.12	2.11	1.74	2.23	3.04	3.03
	HDD10.0	660	206	173	109	19	0	0	0	0	0	19	132	
	HDD18.3	2202	464	406	364	227	91	9	0	1	49	205	387	
	CDD10.0	2011	1	1	4	43	171	285	407	465	347	220	64	3
	CDD18.3	512	0	0	0	0	3	44	149	207	97	11	0	0
	CDH23.3	2139	0	0	0	1	8	34	592	1351	152	2	0	0
	CDH26.7	342	0	0	0	0	0	0	77	260	6	0	0	0
Wind	WSAvg	5.3	6.6	6.3	5.8	5.2	4.6	3.8	4.7	4.2	4.2	5.1	5.8	6.7
Precipitation	PrecAvg	1132	23	33	53	90	97	153	217	217	114	52	51	32
	PrecMax	1654	50	85	124	217	211	366	351	623	262	146	136	87
	PrecMin	716	6	4	7	20	19	47	83	40	14	2	3	3
	PrecStd	221	13	22	30	50	47	83	75	152	68	39	36	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	13.2	15.2	19.8	24.2	25.5	30.3	31.6	27.5	23.5	19.4	14.9
		MCWB	10.5	11.6	11.8	13.4	16.4	21.2	26.6	27.6	24.3	20.3	17.0	13.1
	2%	DB	9.8	11.1	13.2	17.1	21.9	24.2	28.6	30.1	26.0	21.8	17.8	12.7
		MCWB	7.8	9.3	10.6	12.9	16.2	20.7	26.0	26.9	23.1	19.2	16.1	10.7
	5%	DB	8.1	9.2	11.9	15.7	20.3	23.3	27.3	29.2	25.0	20.7	16.7	11.3
		MCWB	6.3	7.1	9.7	12.3	15.8	20.3	25.3	26.5	22.6	18.1	15.0	9.2
	10%	DB	6.8	7.7	10.7	14.5	19.0	22.4	26.2	28.1	24.1	19.8	15.6	9.9
		MCWB	4.6	5.8	8.5	11.7	15.1	20.0	24.6	25.8	21.8	17.4	13.7	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	12.4	12.9	15.7	19.2	23.2	27.8	29.0	25.2	21.9	17.6	13.1
		MCDB	11.8	12.9	13.9	17.4	20.9	24.2	29.0	29.8	26.5	22.4	18.5	14.2
	2%	WB	8.4	9.9	11.5	14.5	17.9	22.0	26.5	27.5	24.1	20.0	16.6	11.2
		MCDB	9.5	10.7	12.6	15.9	19.8	23.1	28.1	29.3	25.2	21.0	17.6	12.4
	5%	WB	6.6	7.6	10.3	13.5	17.0	21.2	25.5	26.8	23.3	19.0	15.4	9.6
		MCDB	7.9	8.9	11.7	14.9	18.7	22.3	26.9	28.7	24.4	20.1	16.4	11.1
	10%	WB	5.1	6.0	8.9	12.4	16.3	20.6	24.8	26.1	22.4	18.0	14.2	8.0
		MCDB	6.8	7.5	10.3	13.9	18.0	21.7	25.9	27.7	23.5	19.4	15.3	9.6
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.7	4.8	5.3	5.5	4.7	4.1	4.3	3.6	3.4	3.5	3.4
		MCDBR	5.1	6.7	7.3	7.0	7.3	5.8	5.3	5.6	4.7	4.5	4.8	5.3
		MCWBR	4.8	5.9	6.0	5.1	4.4	3.6	3.2	3.2	3.3	3.5	4.4	5.1
	5% WB	MCDBR	5.1	6.5	7.0	6.2	6.0	4.8	5.0	5.5	4.6	4.2	4.7	5.2
		MCWBR	5.0	6.2	6.1	5.3	4.3	3.7	3.4	3.5	3.6	3.7	4.7	5.5
Clear-Sky Solar Irradiance	taub	0.460	0.502	0.558	0.571	0.587	0.648	0.617	0.557	0.500	0.483	0.476	0.459	
	taud	1.940	1.869	1.758	1.783	1.802	1.752	1.876	2.041	2.120	2.023	1.972	1.945	
	Ebn at Noon	700	720	718	737	731	686	704	741	764	732	679	668	
	Edh at Noon	147	176	213	218	216	227	200	167	147	149	142	139	
All-Sky Solar Radiation	RadAvg	2.12	2.99	4.33	4.97	5.34	4.81	4.68	4.99	4.35	3.71	2.51	1.86	
	RadStd	0.20	0.33	0.38	0.38	0.55	0.44	0.61	0.68	0.39	0.33	0.28	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+81
Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page